

2.4 Non-linear Graphs

AQA GCSE Mathematics 8300 — Algebra — spec refs A11–A16 — Papers 1 (non-calculator), 2 and 3 (calculator)

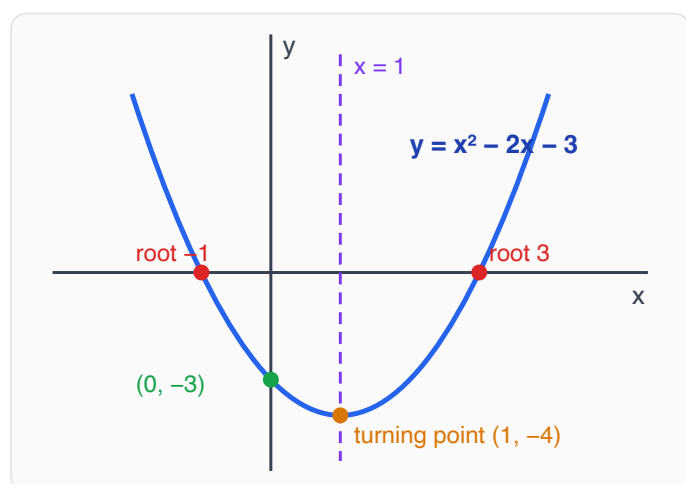
This topic covers quadratic graphs and their roots, intercepts and turning points (A11), recognising and sketching linear, quadratic, cubic and reciprocal graphs (A12), real-life graphs including kinematics (A14). At Higher tier it adds turning points by completing the square, exponential and trigonometric graphs (A12), translations and reflections of graphs (A13), gradients and areas under graphs (A15) and circles centred at the origin with their tangents (A16).

2.4.1 Quadratic Graphs

A **quadratic** graph has equation $y = ax^2 + bx + c$ ($a \neq 0$). Its shape is a **parabola**:

- $a > 0$: U-shaped, with a **minimum** turning point.
- $a < 0$: n-shaped, with a **maximum** turning point.

Every parabola is **symmetrical** about a vertical line through its turning point.



Feature	How to find it	For $y = x^2 - 2x - 3$
y -intercept	Put $x = 0$	$(0, -3)$
Roots (x -intercepts)	Put $y = 0$ and solve	$(x - 3)(x + 1) = 0 \Rightarrow x = 3, x = -1$
Line of symmetry	Halfway between the roots	$x = \frac{-1+3}{2} = 1$
Turning point	Substitute the line of symmetry	$y = 1 - 2 - 3 = -4$, so $(1, -4)$

Plotting a Quadratic

Make a table of values, plot the points and join them with a **smooth curve** (not straight segments). The values should be symmetrical about the turning point.

x	-2	-1	0	1	2	3	4
$y = x^2 - 2x - 3$	5	0	-3	-4	-3	0	5

Exam Tip: Squaring a negative gives a positive: when $x = -2$, $x^2 = 4$, not -4 . Use brackets on your calculator: $(-2)^2$. If your table is not symmetrical, check for this error.

Deducing Roots Algebraically

The roots of $y = ax^2 + bx + c$ are the solutions of $ax^2 + bx + c = 0$. Factorise (or use another method) and set each bracket to zero.

Worked Example: Find the roots and turning point of $y = x^2 + 2x - 8$.

$$(x + 4)(x - 2) = 0 \Rightarrow x = -4 \text{ or } x = 2$$

$$\text{Line of symmetry: } x = \frac{-4+2}{2} = -1, \quad y = (-1)^2 + 2(-1) - 8 = -9$$

$$\text{Turning point } (-1, -9)$$

Worked Example: A ball's height is $h = 20t - 5t^2$ metres. When does it land?

$20t - 5t^2 = 0 \Rightarrow 5t(4 - t) = 0 \Rightarrow t = 0$ or $t = 4$. It lands after **4 seconds**. By symmetry its greatest height is at $t = 2$: $h = 40 - 20 = 20$ m.

Solving Equations Graphically

To solve $f(x) = k$ using the graph of $y = f(x)$, draw the horizontal line $y = k$ and read off the x -coordinates where the line meets the curve. The roots (solutions of $f(x) = 0$) are where the curve meets $y = 0$.

Worked Example: Use the table above to solve $x^2 - 2x - 3 = 5$.

Draw $y = 5$. It meets the curve at $x = -2$ and $x = 4$.

2.4.2 Turning Points by Completing the Square Higher

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

If $y = (x + p)^2 + q$, the turning point is $(-p, q)$ and it is a **minimum** (since $(x + p)^2 \geq 0$).

Worked Example: Find the turning point of $y = x^2 - 8x + 20$.

$$x^2 - 8x + 20 = (x - 4)^2 - 16 + 20 = (x - 4)^2 + 4$$

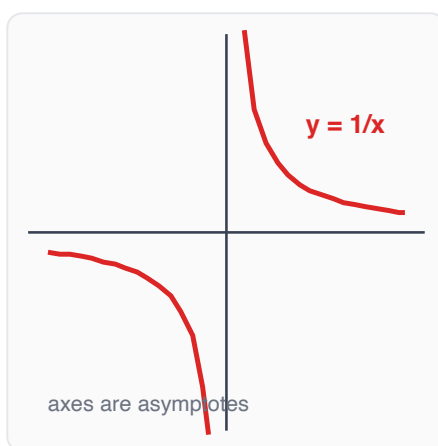
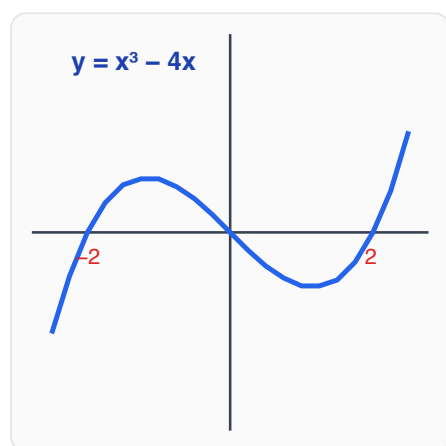
Turning point **(4, 4)** (minimum). Since the minimum value 4 is above zero, the graph has **no real roots**.

Worked Example: Write $x^2 + 6x + 2$ in the form $(x + p)^2 + q$ and state the turning point.

$(x + 3)^2 - 9 + 2 = (x + 3)^2 - 7$, so the turning point is $(-3, -7)$.

Exam Tip: The sign of the x -coordinate flips: $(x - 4)^2$ gives $x = 4$; $(x + 3)^2$ gives $x = -3$. The y -coordinate keeps its sign.

2.4.3 Cubic and Reciprocal Graphs



Type	General form	Shape and key features
Linear	$y = mx + c$	Straight line
Quadratic	$y = ax^2 + bx + c$	Parabola (U or n), one turning point, symmetrical
Cubic	$y = ax^3 + \dots$	S-shaped; $y = x^3$ passes through the origin; can have 1, 2 or 3 roots. If $a > 0$ it goes from bottom-left to top-right.
Reciprocal	$y = \frac{k}{x}, x \neq 0$	Two separate branches; the axes are asymptotes . For $k > 0$, branches in the 1st and 3rd quadrants; symmetrical about $y = x$ and $y = -x$.

An **asymptote** is a line that a graph gets closer and closer to but never meets. For $y = \frac{1}{x}$: as x gets very large, y gets closer to 0; $x = 0$ is not allowed because you cannot divide by zero.

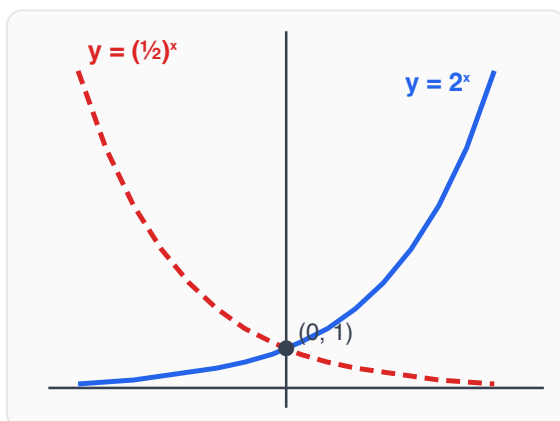
Worked Example: Find the roots of $y = x^3 - 4x$.

$x(x^2 - 4) = x(x - 2)(x + 2) = 0 \Rightarrow x = -2, 0, 2$. It is a cubic with three roots (see the graph above).

2.4.4 Exponential and Trigonometric Graphs

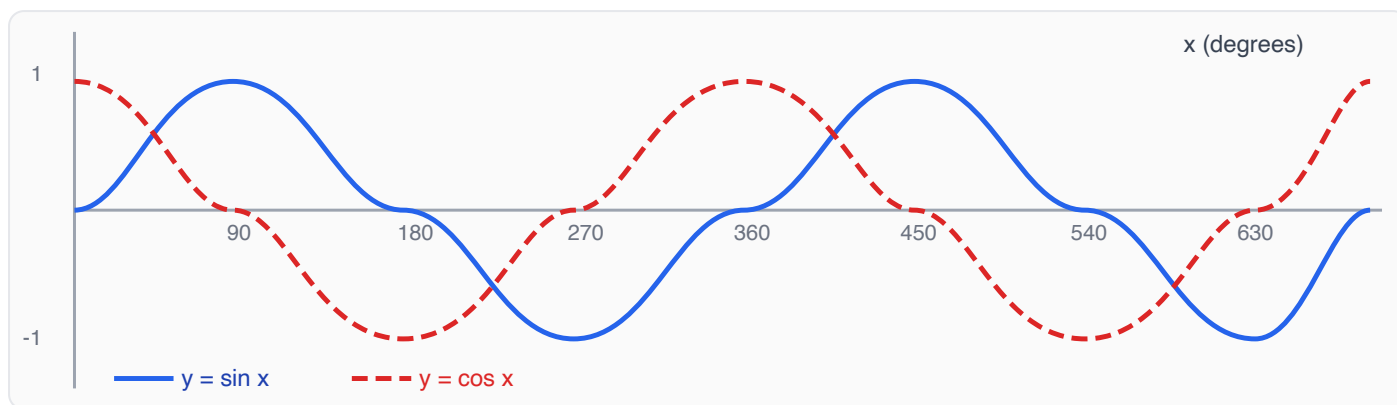
Higher

Exponential Graphs $y = k^x$



- Every graph $y = k^x$ ($k > 0$) passes through $(0, 1)$, because $k^0 = 1$.
- y is always positive: the x -axis is an asymptote.
- $k > 1$: exponential **growth** (increasing ever more steeply). $0 < k < 1$: exponential **decay**.
- Real-life models: compound interest $V = P \times 1.04^t$, population growth, depreciation. The vertical intercept is the starting value P .

Trigonometric Graphs



Graph	Period	Range	Key points (0° to 360°)
$y = \sin x$	360°	$-1 \leq y \leq 1$	$(0, 0)$, max $(90^\circ, 1)$, $(180^\circ, 0)$, min $(270^\circ, -1)$, $(360^\circ, 0)$
$y = \cos x$	360°	$-1 \leq y \leq 1$	max $(0, 1)$, $(90^\circ, 0)$, min $(180^\circ, -1)$, $(270^\circ, 0)$, max $(360^\circ, 1)$
$y = \tan x$	180°	all values	$(0, 0)$, $(45^\circ, 1)$, asymptotes at $x = 90^\circ$ and $x = 270^\circ$, $(180^\circ, 0)$

Worked Example: How many solutions does $\sin x = 0.5$ have for $0^\circ \leq x \leq 360^\circ$?

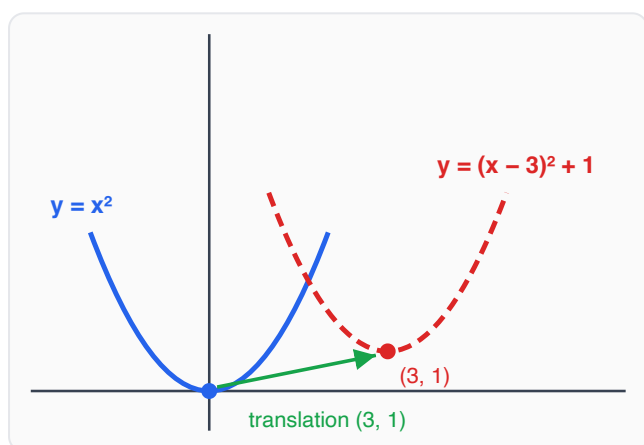
The line $y = 0.5$ meets the sine curve twice: $x = 30^\circ$ and, by symmetry about $x = 90^\circ$, $x = 180^\circ - 30^\circ = 150^\circ$. So **2 solutions**.

Exam Tip: The graphs repeat for angles of any size, including negative angles — use the period and symmetry to find further solutions rather than relying only on your calculator, which gives one answer.

2.4.5 Translations and Reflections of Graphs

Higher

New graph	Transformation of $y = f(x)$	Point (x, y) moves to
$y = f(x) + a$	Translation $\begin{pmatrix} 0 \\ a \end{pmatrix}$ (a up)	$(x, y + a)$
$y = f(x + a)$	Translation $\begin{pmatrix} -a \\ 0 \end{pmatrix}$ (a left)	$(x - a, y)$
$y = -f(x)$	Reflection in the x -axis	$(x, -y)$
$y = f(-x)$	Reflection in the y -axis	$(-x, y)$



Worked Example: $y = f(x)$ has a maximum at $(1, 6)$ and crosses the x -axis at $(-2, 0)$ and $(4, 0)$.

$y = f(x + 2)$: shift 2 left $\Rightarrow$ maximum $(-1, 6)$, roots $(-4, 0)$ and $(2, 0)$.

$y = -f(x)$: reflect in the x -axis $\Rightarrow$ **minimum** $(1, -6)$, roots unchanged.

$y = f(x) - 3$: shift 3 down $\Rightarrow$ maximum $(1, 3)$.

Exam Tip: Changes *inside* the bracket affect x and work the "opposite way": $f(x - 4)$ moves the graph 4 to the **right**. Changes *outside* affect y and work as expected.

2.4.6 Real-Life and Kinematic Graphs

Distance–time graph: gradient = **speed**; horizontal line = stationary; steeper = faster; a line sloping back down = returning towards the start.

Velocity (speed)–time graph: gradient = **acceleration**; horizontal line = constant speed; negative gradient = deceleration; area under the graph = **distance travelled**.

Worked Example: A train travels 15 km in its first 20 minutes, stops for 10 minutes, then returns 15 km in 30 minutes. Find the speed in each moving section.

First section: $15 \div \frac{20}{60} = 45 \text{ km/h}$. Return: $15 \div \frac{30}{60} = 30 \text{ km/h}$.

Worked Example: A car's speed increases uniformly from 0 to 24 m/s in 8 s. Acceleration
 $= \frac{24}{8} = 3 \text{ m/s}^2$.

Filling Containers and Other Contexts

When a container is filled at a constant rate, the depth–time graph is a straight line for any part with a constant cross-section. Wider parts fill more slowly (shallower gradient); narrower parts fill faster (steeper gradient). A container with sloping sides gives a curve.

Reciprocal and Exponential Models

Inverse relationships give reciprocal graphs: if n workers take $t = \frac{24}{n}$ hours, the graph is a reciprocal curve that never reaches the n -axis. At Higher, growth and decay ($V = 500 \times 1.04^t$) give exponential graphs; read approximate solutions from the graph, e.g. when the value first exceeds £600.

2.4.7 Gradients and Areas Under Graphs

Higher

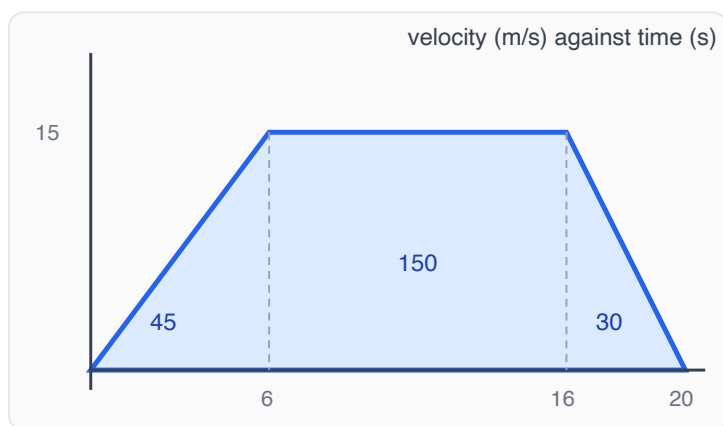
Gradient of a Curve

- **Instantaneous rate of change** at a point: draw a **tangent** to the curve at that point and find its gradient.
- **Average rate of change** between two points: find the gradient of the **chord** joining them.
- On a distance–time curve the tangent gives the speed at that instant; on a velocity–time curve it gives the acceleration.

Area Under a Graph

Split the area into triangles, rectangles and trapezia. For a curve, use **trapezia of equal width** to estimate the area.

$$\text{Area of a trapezium strip} = \frac{1}{2} \times \text{width} \times (y_{\text{left}} + y_{\text{right}})$$



Worked Example: A car accelerates from rest to 15 m/s in 6 s, travels at 15 m/s for 10 s, then decelerates to rest in 4 s. Find the total distance.

$$\frac{1}{2} \times 6 \times 15 + 10 \times 15 + \frac{1}{2} \times 4 \times 15 = 45 + 150 + 30 = \mathbf{225 \text{ m}}$$

Worked Example: A velocity–time curve passes through (0, 0), (2, 6), (4, 10), (6, 12). Estimate the distance in the first 6 s.

$$\frac{1}{2}(2)(0 + 6) + \frac{1}{2}(2)(6 + 10) + \frac{1}{2}(2)(10 + 12) = 6 + 16 + 22 = \mathbf{44 \text{ m}}$$

The curve gets less steep, so it bends above the trapezium tops: this is an **underestimate**.

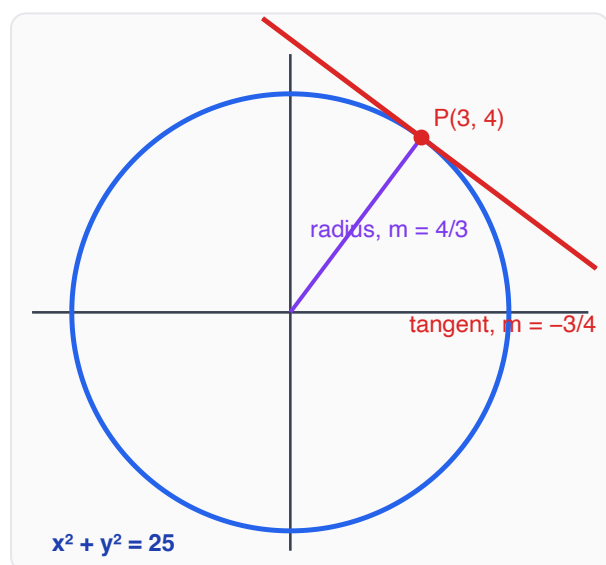
Exam Tip: Always state units: the area under a velocity–time graph ($\text{m/s} \times \text{s}$) is in metres; the gradient ($\text{m/s} \div \text{s}$) is in m/s^2 . In financial graphs the gradient is a rate such as £ per year.

2.4.8 Equation of a Circle and Tangents Higher

The circle with centre the origin and radius r has equation

$$x^2 + y^2 = r^2$$

e.g. $x^2 + y^2 = 49$ has centre $(0, 0)$ and radius 7. A point lies on the circle if its coordinates satisfy the equation.



Tangent at a point P : the tangent is perpendicular to the radius OP .

1. Gradient of radius $OP = \frac{y_P}{x_P}$.
2. Tangent gradient = negative reciprocal.
3. Substitute P into $y = mx + c$ to find c .

Worked Example: Find the tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

Radius gradient $= \frac{4}{3}$, so tangent gradient $= -\frac{3}{4}$.

$$4 = -\frac{3}{4}(3) + c \Rightarrow c = 4 + \frac{9}{4} = \frac{25}{4}$$

$$y = -\frac{3}{4}x + \frac{25}{4}, \text{ i.e. } 3x + 4y = 25$$

Worked Example: Find the tangent to $x^2 + y^2 = 20$ at $(2, -4)$.

Check: $4 + 16 = 20$ ✓. Radius gradient $= \frac{-4}{2} = -2$, so tangent gradient $= \frac{1}{2}$.

$$-4 = \frac{1}{2}(2) + c \Rightarrow c = -5 \Rightarrow y = \frac{1}{2}x - 5$$

Exam Tip: The radius is $\sqrt{r^2}$: for $x^2 + y^2 = 50$ the radius is $\sqrt{50} = 5\sqrt{2}$, not 25. To find the equation from a point on the circle, use $r^2 = x^2 + y^2$, e.g. through $(6, -8)$: $r^2 = 36 + 64 = 100$.

Quick Reference Summary

Idea	Key fact
Roots	Solve $y = 0$; line of symmetry halfway between roots
Turning point Higher	$(x + p)^2 + q \Rightarrow (-p, q)$
Graph shapes	Quadratic U/n, cubic S, reciprocal two branches
Exponential Higher	$y = k^x$ through $(0, 1)$
Trig Higher	sin, cos period 360° ; tan period 180°
Transformations Higher	$f(x) + a$ up; $f(x + a)$ left; $-f(x)$ and $f(-x)$ reflect
Kinematics	D-T gradient = speed; V-T gradient = acceleration, area = distance

Circle

Higher

$$x^2 + y^2 = r^2; \text{tangent} \perp \text{radius}$$